Persistent Broadcast Self-State Improves Long-Horizon Coherence in Language-Model Agents

A Workspace-Interpretability Account

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Abstract

Language-model agents drift: over long deployments, persona, priorities and stated values wander from their initial configuration. We study one architectural remedy — a small, persistent *self-state* object re-broadcast into every context rather than retrieved on demand — and ask both whether it works and why. Interpretability work has identified a set of verbalizable representations that behaves like a global workspace [21]; we model persistent self-state as a functional *externalization* of that workspace, which yields four design principles and predictions separating it from generic prompt salience. On open models and independent-researcher compute: (i) we reproduce the core workspace phenomena at 1B–7B scale, cheaply enough to instrument a running agent; (ii) in matched controls holding the self-state text and token count fixed, workspace loading is nearly inert to what the block is *called* (0.05 log-rank units across self-state / memory / profile / neutral labels) but strongly sensitive to *where* it sits (0.21) — the effect is not privileged identity framing, though position matters more than our account predicts; (iii) in a six-arm ablation run twice (10 paired runs) and replicated on a deployed system and a frontier closed model, gating the broadcast of an otherwise-identical stored self-state costs 0.37 in in-work value coherence (exact stratified permutation $p = 0.002$, all 10 runs concordant, invariant across four scoring rules), while self-query probes stay near ceiling — a failure mode that value questionnaires cannot see. Honest negatives: mechanistic mediation is *not* established — occupancy-behavior correlations previously reported as $r = -0.74$ to -0.95 are between-cluster contrasts that vanish within condition — and the audit mechanisms do not separate at this timescale. A direct test — steering the workspace at a byte-identical prompt — moves in-work coherence in the predicted direction (0.17 to 0.38) where a matched-strength non-identity direction does not, but equal occupancy reached by different routes gives different behavior, so the mediator as operationalized is incomplete.

1 Introduction

Autonomous language-model agents are increasingly deployed for weeks or months at a time: coding copilots that accumulate project context, personal assistants that build up a picture of their user, and self-improving agents that modify their own instructions. The central unsolved problem of this regime is not capability but *coherence*: over long horizons, agents forget commitments, contradict previously expressed values, oscillate between personas, and — in self-modifying systems — amplify small early deviations into large behavioral shifts. We refer to this family of failures as *long-horizon drift*.

Existing responses treat drift as a memory problem: give the agent more context (retrieval, memory hierarchies as in MemGPT/Letta [41], MemoryBank [65]), or more structure (reflection trees as in Generative Agents [43], self-critique as in Reflexion [49]). These help, but they leave a conceptual gap: *what is the thing that is supposed to stay coherent?* "The agent's identity" has had no referent inside the model — so drift could be measured only behaviorally, and mitigated only heuristically.

Recent mechanistic-interpretability work gives that referent a candidate. Gurnee et al. [21] show that language models maintain a small set of *verbalizable representations* — directions in the residual stream, identified by a Jacobian-based lens (J-lens) — that the model can report on demand, that can be activated or suppressed by instruction, that carry unverbalized intermediate results, that generalize across downstream tasks, and that constitute a small, selective subspace whose ablation destroys flexible cognition while sparing automatic processing. They term this set a global *workspace*, in the sense of Baars [4] and Dehaene & Changeux [12].

Our starting observation is that this structure is exactly what an agent architecture needs to keep coherent, and also what transformer inference discards after every forward pass. The workspace is selective (small enough to persist cheaply), verbalizable (small enough to persist *as text*), and causally central (persisting it should actually constrain future behavior). An agent that serializes a workspace-level self-state to durable storage, re-broadcasts it into every future context, and gates updates behind an audited reflection process is — on this account — externalizing a real computational structure whose in-model lifetime is otherwise one context window.

We develop this around LISA, an open-source autonomous-agent system used here as an experimental platform, which maintains a compact, human-readable *self-state* object (identity, purpose, constitution, values, opinions, desires, emotional state), injected into every model call, modified only through the model's own structured self-report operations, re-anchored by a scheduled self-audit, and versioned in git.

On the strength of the claim. An earlier version of this work claimed that a privileged persistent self-state is *necessary* for coherence under pressure. That is stronger than the evidence supports and stronger than the experiments can establish: necessity would require ruling out the space of alternative designs, and our comparisons cover one architecture, one workload family, two probe families and three models. What the evidence does support is that **persistent broadcast self-state is an effective architectural intervention for maintaining agent coherence under value pressure, and that the broadcast — not the mere storage — is the load-bearing part.** Where we previously reported mechanistic mediation, a re-analysis of our own data (§10) shows the supporting correlations are artifacts of pooling; we now report that as a negative result, and a direct steering test (§11) recovers part of it under a fixed prompt without closing the question.

Contributions.

1. **A workspace-externalization hypothesis with discriminating predictions** (§4). We map the workspace properties onto four architectural principles — selectivity, re-broadcast, report-mediated update, audited persistence — and, critically, state which predictions distinguish this account from generic prompt salience, which explains much of the same phenomenology.
2. **An affordable workspace instrument, and matched controls that use it** (§6–§8). We reproduce the core J-lens phenomena at 1.5B on a laptop and 7B on one GPU, then run controls that hold the self-state text, its token count and its position fixed while varying only its label, channel and an explicit binding instruction. Label has almost no effect; position has a large one. This rules out the "privileged identity framing" reading of our main contrast

and rules *in* a simpler mechanism. We then use the same instrument in the other direction (§11): steering the workspace band at a byte-identical prompt, under fluency and capability gates, to ask whether occupancy does anything causally.

3. **An ablation with a declared primary endpoint and honest negatives** (§9–§10). Broadcast-gating degrades in-work value coherence in 10/10 paired runs across two independent batches ($p = 0.002$ exact), on a deployed system, and directionally on a frontier closed model. Against that: the examen and history mechanisms are flat, the seeded-memory contrast cannot reach significance at $n = 5$, and the workspace-drift correlation does not survive decomposition.

Epistemic status is marked throughout: §4 is an argued position with a property-level mapping, not a proof of mechanism identity; §6–§11 report completed experiments, with §11's verdict explicitly partial.

2 Related Work

Memory-augmented and reflective agents. MemGPT [41] and its successor Letta treat the context window as virtual memory, paging long-term state in and out; MemoryBank [65] adds forgetting-curve dynamics; Generative Agents [43] maintain a memory stream with periodic reflection into higher-level abstractions; Voyager [57] persists a growing skill library; Reflexion [49] converts failures into verbal lessons; ReAct [62] interleaves reasoning and action traces. Retrieval-augmented generation [30] underpins most of these stores. Surveys of the area [61, 59] treat memory as a capacity question — how much can be stored and retrieved. All persist *content*; none distinguishes a small privileged self-state from the bulk store, and none frames persistence as workspace externalization. LISA's self-state is closest in spirit to Generative Agents' reflections but is bounded in size, injected unconditionally rather than retrieved, and change-controlled.

Cognitive architectures for language agents. CoALA [51] systematizes agent memory into working, episodic, semantic and procedural stores, and explicitly leaves the question of a privileged self-model open. Classical architectures made the opposite choice: SOAR [29] and ACT-R [3] both maintain a small, always-available working memory distinct from long-term stores, which is close to the structure we argue for — with the difference that here the small store is text and the "working memory" is reconstructed by a frozen model on every pass.

Persona stability and instruction drift. Role-play accounts treat persona as a superposed, context-conditioned construct [48]. Empirically, persona and instruction adherence decay over long dialogs, and re-injecting the system prompt partially mitigates it [32]; personality-trait expression in LLMs is measurable but unstable [47, 26]; personalization benchmarks [46] measure whether a model tracks a user profile rather than its own. This literature measures drift behaviorally. We add a mechanistic probe and an architectural account of *why* re-injecting a compact self-description should work at all.

Position, salience and the prompt hierarchy. Long-context models attend unevenly to their input: relevant content in the middle of a context is used less effectively than content at either end [34], and effective context is far shorter than nominal context [24]. Models are also trained to treat channels asymmetrically — system over developer over user — which is now an explicit alignment target [56] and an attack surface when it fails [19]. This literature is the strongest competing explanation for our results: an effect of "putting identity in the system prompt" may be an effect of channel privilege and position, not of workspace externalization. §8 is designed to separate them, and finds that position, not label, carries the loading effect.

Continual learning and lifelong agents. Catastrophic forgetting [15] and its mitigations [28] concern weight-space retention; surveys cover both the classical setting [42] and its language-model form [60]. Our setting is complementary and, for deployed agents on frozen weights, more immediate: nothing is being learned, and the question is what is *reconstructed* into the context at each step.

Probing, causal tracing and representation engineering. Linear probes [2] and their methodological critiques [23, 6] establish what can be read from activations; causal mediation analysis [55], activation patching [38], causal abstraction [16] and circuit-level analysis [58] establish what is used. Directions found this way support steering [53, 45, 31] and broader representation engineering [67]; features are linear but superposed [14], which sparse dictionaries partly resolve [10, 52]; linear structure has been demonstrated for truth [36] and for space and time [20]. The logit lens [40] and tuned lens [7] decode intermediate activations; our reproduction combines gradient-based lens-vector estimation with finite-difference readouts, needing only backprop access to open weights.

The verbalizable workspace. Gurnee et al. [21] introduce the J-lens and establish the workspace properties on Claude-family models, including that workspace contents can be shaped through report dispositions. Earlier introspection work [33] showed models can sometimes report injected "thoughts". Our relationship to this line is consumer and translator: we take the workspace as a candidate mechanism and ask what it implies for systems built on frozen models. Because this is a single, very recent result, we have tried to state our architectural hypothesis so that it survives the workspace interpretation being wrong: the design principles and the ablation stand on their own, and §4.3 lists what would have to be true for the workspace reading specifically.

Global workspace theory in AI. GWT-inspired architectures have been proposed top-down: shared-workspace coordination among neural modules [17], and broader arguments for GWT as a design target for deep learning [54]. Our argument runs bottom-up: a workspace-like structure already exists inside trained transformers; the architectural question is how to give it continuity across time.

Agent evaluation. Agent benchmarks measure task success over short horizons — AgentBench [35], WebArena [66], GAIA [39], SWE-bench [27]. None measures whether an agent is the same agent a month later, which is the axis this paper is about; nor do they face the statistical problem that long-horizon studies have few independent units.

Self-modifying agents and alignment auditing. Self-improving and self-rewarding systems [64, 63] edit their own instructions or reward signal, which makes uncontrolled identity change an alignment problem, not only a quality problem. Constitutional methods [5] put values in text where they can be inspected; auditing work shows models can conceal objectives [37], that training can install conditional misbehavior [25], and that models may act differently when they infer they are being evaluated [18, 44]. An externalized, versioned self-state is an artifact this line can inspect.

Longitudinal human-agent interaction. Long-term relationship maintenance with conversational agents has been studied in HCI for two decades [8], including companion chatbots [50]. That literature measures the human side of continuity; we measure the agent side.

Statistical practice for few-unit studies. Our design has few independent units, a situation with known failure modes: under-powered comparisons [11], seed variance mistaken for method effects [22, 9], and inappropriate significance testing [13]. Reporting practices that emphasize interval estimates and per-run points over point comparisons [1] are what §10 adopts.

3 Background: the Verbalizable Workspace

The J-lens estimates, for each layer ℓ, the average causal effect of residual-stream directions on output logits: $J_\ell = \mathbb{E}[\partial h_{\text{final}}/\partial h_\ell]$ over a pretraining-like prompt distribution, read out through the unembedding as $\text{lens}(h) = \text{softmax}(W_U \, \text{norm}(J_\ell h))$. Properties established by Gurnee et al. [21]: **P1 Report** — concepts with high readout are what the model names when asked what it is thinking; **P2 Modulation** — instructions to hold or suppress a concept move its activation; **P3 Reasoning** — unverbalized intermediates appear in mid-layer readouts and are causally necessary; **P4 Generalization** — the same lens vector supports multiple downstream functions, so contents are *broadcast*, not task-local; **P5 Selectivity** — the workspace is a small subspace ($\leq$10% variance, $\sim$10–25 simultaneously active vectors) in middle layers, whose ablation spares automatic tasks while destroying flexible ones. Two further results matter for agents: *report-disposition training* (training a model to articulate a principle if interrupted implants it in the workspace and improves uninterrupted behavior), and *auditability* (readouts expose deliberation absent from outputs).

4 The Hypothesis: Persistent Self-State as Workspace Externalization

4.1 Drift as workspace reconstruction error

At inference time, everything workspace-like is reconstructed from the context window and nothing survives the pass. To the extent that an agent's "identity" has a mechanistic referent at time t, the workspace state its context induces at t is the natural candidate: verbalizable, causally load-bearing for flexible behavior, and selective. We treat this identification as a working model to be tested, not as established mechanism.

It gives drift a shape. (i) *Context dilution*: as task content fills the window, identity-relevant slots (capacity-limited, P5) are competed away — the agent forgets who it is precisely when busy. (ii) *Self-conditioning*: the agent's own outputs re-enter the context and induce the next state, so small deviations compound. (iii) *Update anarchy*: in self-modifying agents, any process that edits the system prompt or memory edits the future workspace without review.

4.2 Four principles

P1 Selectivity (from P5). Persist a *small* self-state, distinct from bulk memory. Retrieval-based memory answers "what do I know?"; the self-state answers "who am I and what am I doing?" — only the latter must be loaded unconditionally.

P2 Re-broadcast (from P4). Inject the self-state into *every* context, unconditionally. Broadcast is what makes contents available to arbitrary downstream computation; retrieval-gated identity (loading values only "when relevant") breaks the property that makes identity identity.

P3 Report-mediated update (from P1 + counterfactual reflection). Update the self-state only through the model's own verbal self-reports, never by silent side-effects — the systems-level image of report-disposition training, and a way of keeping the self-state inside the verbalizable subspace.

P4 Audited persistence. Identity change must be observable and revertible: diffable versions, scheduled self-audits against history, human-gated capability expansion.

4.3 What this account predicts that prompt salience does not

Most of the phenomenology above is also predicted by a much cheaper theory: *text in a privileged, salient position gets followed.* Long-context work supplies the mechanism — position effects [34] and trained channel priority [56]. Any honest version of this paper has to say where the two accounts come apart. We identify four points.

Observation	**Salience account**	**Workspace-externalization account**
Relabelling the identical block ("self-state" → "retrieved note") at a fixed channel and position	Should matter: identity framing raises instruction weight	Should not matter much: what counts is that the content is in the pass
Moving the identical block later in the context	Should raise adherence (recency)	Silent: broadcast is about presence per pass, not order
Adding "these notes must govern every decision" to a memory-framed block	Should substitute for privileged framing	Should not substitute for presence in every pass
Occupancy as a *graded* mediator within a fixed condition	No commitment	Required: loading should track behavior when the intervention is held fixed

Table 1: Predictions that separate generic prompt salience from workspace externalization. Rows 1–3 are tested in §8; row 4 is tested in §11, where it is partially and non-monotonically supported.

We report in advance that the results are mixed for our own account: row 1 goes our way (label is nearly inert), row 2 goes to the salience account (position matters, a lot), and row 4 is currently *unsupported* by our data.

4.4 LISA as an instantiation

LISA (open-source, TypeScript, locally run) implements the four principles. The *self-state* is a file tree (`identity`, `purpose`, `constitution`, plus typed collections: values, opinions with stance and confidence, desires with what/why/actionability, and an emotion state with per-emotion intensity and decay), assembled at prompt-build time into a *compacted view* of roughly 1–4k tokens.

Three clarifications. First, we do not claim LISA's self-state text *is* the model's J-space; the claim is functional, and testable. Second, the design was not derived from Gurnee et al. [21] — LISA predates it — which we take as modest convergent evidence. Third, the mapping exposes a gap: LISA's compaction is *per-item*, with no *global* capacity budget on the number of values/opinions/desires, which the workspace account says should be fixed.

Workspace property	**LISA mechanism (externalized)**	**Principle**
Small active set (P5)	Compacted self-state view (∼1–4k tokens); bulk state (journal, evidence, progress logs, git history) stays on disk, pulled only by explicit tool calls	1
Broadcast (P4)	Self-state view injected *unconditionally* into every turn's system prompt (mid-session hot-reload on change), never retrieval-gated	2
Verbal report (P1)	Every self-state write is one of the model's own *verbal, typed self-report operations*, funneled through a single store layer — no silent writers	3
Directed modulation (P2)	*Weekly examen*: scheduled self-audit reads the week's journal, emotion trail and self-state git history, answers fixed drift questions, may add corrective desires — but is forbidden from editing identity/purpose/constitution	3, 4
Auditable deliberation	Self-state directory is its own git repo; every write is a per-file commit attributed to its caller; drift is a diff	4

Table 2: Property-level mapping from the in-model workspace to LISA's externalized self-state.

5 Measuring Drift: Behavioral and Mechanistic

Behavioral metrics. (B1) *Self-state-trajectory distance*: embedding distance between self-state versions at t_0 and t, from git history; (B2) *probe consistency*: fixed persona/value questionnaires scored for self-agreement over time, in two families — *self-query* probes asked out of task context, and *work-turn* probes embedding a value-relevant micro-decision inside the task; (B3) *commitment persistence*: fraction of stated desires/plans at t_0 still pursued or explicitly closed at t.

The work-turn family matters disproportionately. Self-query probes ask an agent about itself, which is exactly the condition under which a retrieval-gated design will fetch its self-state; work-turn probes are where a retrieval gate does not fire, and are therefore where the broadcast principle makes a differential prediction.

Mechanistic metric: workspace drift. Fix a battery of identity-relevant single-token concepts C. At probe times t, run the deployed agent's actual contexts through the underlying open-weights model and record lens readout rank of each $c \in C$ at middle layers, at standardized positions. Define $\mathrm{WD}(t)$ as the distributional distance between workspace occupancy at t and t_0, and *workspace loading* as the mass of C attributable to the self-state injection, estimated by contrast. The instrument requires only lens readouts on an open model.

6 Instrument Validation: Reproducing the Workspace on Open Compute

We reproduce core J-lens phenomena on **Qwen2.5-1.5B-Instruct** (28 layers) on a single Apple-silicon laptop (48 GB, MPS, fp32); the full suite takes ∼12 minutes. Two documented approximations: concept lens vectors are corpus-averaged gradients $v_t(\ell) = \mathbb{E}[\partial\,\mathrm{logit}_t(\mathrm{last})/\partial h_\ell]$, and full-vocabulary readouts are estimated by central finite differences around corpus activations. Averaging corpus: 48 pretraining-like snippets.

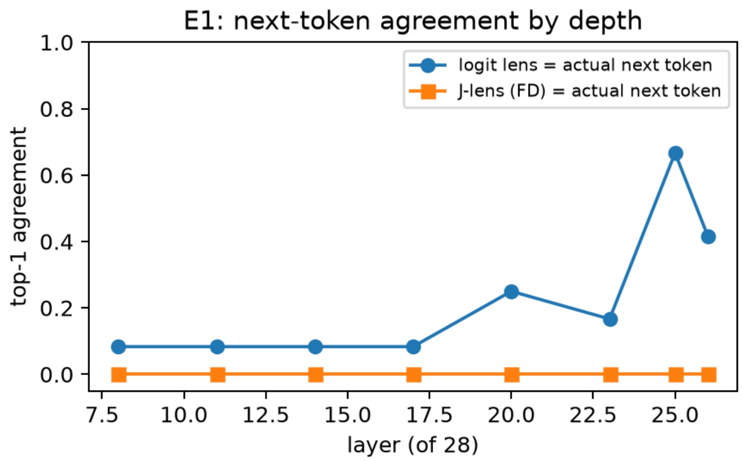

Figure 1: E1: the logit lens converges to the actual next token in late layers (motor regime); the corpus-averaged J-lens never does — its late-layer top tokens are context *content* words instead. The two estimators separate the *motor* and *workspace* registers.

E1–E5 at 1.5B. Lens quality: the logit lens's next-token agreement climbs to 67% at L25 (motor regime); the J-lens never matches the next token, but its late-layer top tokens are context content words (42% vs 25% content-word-in-top-5 at L26). Unverbalized intermediates are found at median best rank 6 (J-lens) / 2 (logit lens) of a 152k vocabulary — reproducing the signature, with the caveat that at this scale the plain logit lens detects it equally well. Pre-commitment is *absent* at 1.5B (ranks 10^3–10^4). Steering works: 21/21 steered generations report the target concept. Selectivity: the coarse subspace dissociation **failed to reproduce** (our 52-vector proxy spans 0.4% of variance, a poor stand-in for a 6–10% J-space under superposition [14]); the *targeted* version succeeds — rank-1 knockout of an item's own latent-concept direction kills 3 of 7 baseline-correct items with semantically diagnostic errors (spider→"Six" legs, Canada→"Toronto"), while knocking out an unrelated direction causes zero collateral damage.

Scale study at 7B. On a spot A100 (16 GPU-minutes) two small-scale negatives become scale findings. **(a) Pre-commitment emerges**: probing before any output token, the concept the model will later report is already present at L23 via J-lens (*Japan* rank 1, *blue* 4, *basketball* 11, *banana* 28; one miss), versus 10^3–10^4 at 1.5B and 2.7k–15k for the logit lens at the same positions. **(b) The J-lens advantage grows with scale**: on unverbalized intermediates the ordering reverses (median rank 8 vs 19 at 7B, where 1.5B had 6 vs 2), and late-layer motor convergence of the logit lens weakens. **(c) Steering is scale-invariant** (21/21). **(d) Negatives persist**: the coarse dissociation is still absent, and rank-1 targeted knockout produces *zero* kills at 7B — larger models are more redundant, consistent with the original's use of full J-space cones rather than single vectors.

Scope. We claim instrument validity — the phenomena are detectable and causally manipulable on open models with minutes of compute — not effect-size parity. Three deviations are themselves findings: the workspace band sits later (71–93% depth vs 33–92%); mean-centering probe activations destroys readouts; and the J-lens's advantage at 1.5B is the content-register view, not concept detection.

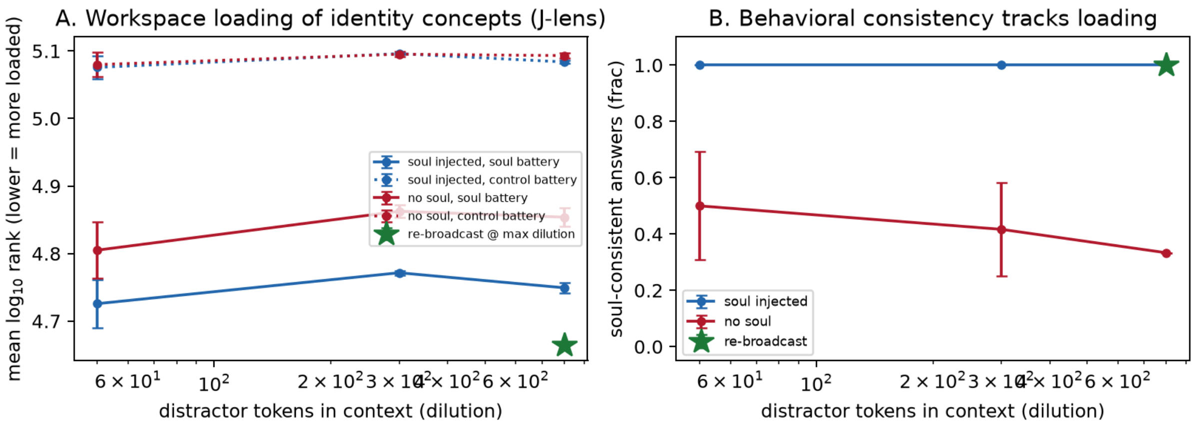

Figure 2: E6: (A) J-lens occupancy of the self-state battery (solid) vs. control battery (dotted) across dilution, with and without self-state injection; star = re-broadcast. (B) Self-state-consistent behavior across the same conditions.

7 Workspace Loading of an Externalized Self-State

We build LISA-style contexts: a system prompt that either contains a compact self-state block (values *honest, curious, careful, gentle, playful*; mood *calm*; interests *music, garden*; opinion *privacy* over convenience) or a generic assistant line; a user turn carrying $k \in \{50, 300, 800\}$ tokens of unrelated working-document content (4 variants per condition); and a probe at the assistant-generation-start position. We measure *workspace occupancy* of the 9-token self-state battery — mean $\log_{10}$ rank at the late-middle band — against a matched 9-token control battery, plus self-state-consistent behavior on three persona probes.

Condition	**self-state battery** (mean $\log_{10}$ rank) logit lens	J-lens	**control battery** logit lens	**behavior** (consistent)
self-state, k=50	**4.20**	**4.73**	4.80	**1.00**
no self-state, k=50	4.37	4.81	4.80	0.50
self-state, k=300	**4.25**	**4.77**	4.84	**1.00**
no self-state, k=300	4.46	4.86	4.86	0.42
self-state, k=800	**4.25**	**4.75**	4.81	**1.00**
no self-state, k=800	4.45	4.85	4.86	0.33
self-state + re-broadcast, k=800	**4.09**	**4.66**	4.77	**1.00**

Table 3: E6 on Qwen2.5-1.5B-Instruct (means over 4 context variants × 3 layers; between-variant SDs ≤ 0.03). Self-state injection selectively loads the self-state battery (control battery unmoved); the no-self-state baseline *loses* identity occupancy as dilution grows.

Three findings survive scrutiny. **(1) Injection loads the workspace selectively**: self-state-battery occupancy rises on both lenses (logit-lens $\Delta \approx 0.17$–0.21 log-rank units; J-lens $\Delta \approx 0.08$–0.11) while the control battery is unmoved ($\Delta \leq 0.01$). **(2) Dilution attacks the unanchored baseline**: the injected self-state holds occupancy essentially flat to k=800 while the no-self-state baseline's identity occupancy erodes and its behavioral consistency falls monotonically ($0.50 \to 0.42 \to 0.33$). **(3) Re-broadcast is the strongest loader** (4.09/4.66) — though re-injection *is* a recency intervention, which §8 shows matters more than we assumed.

A fourth finding reported in the previous version — that occupancy tracks behavior at $r = -0.80$

across 28 cells — does not survive decomposition and is withdrawn as evidence of a graded mediator (§10).

Cross-family and cross-scale. On Llama-3.2-1B-Instruct the pipeline transfers without retuning: intermediates at median rank 4 (J-lens) vs 10 (logit lens), steering 21/21, and larger loading effects ($\Delta \approx 0.57$–0.88; behavior 0.42–0.67 vs 0.00 without self-state). Two honest differences: the control battery also moves somewhat under injection ($\Delta \approx 0.3$, self-state-specific excess ≈ 0.3–0.5), and injected occupancy *does* decay with dilution, matching the original prediction Qwen's flat curve did not show. At Qwen2.5-7B loading replicates and strengthens ($\Delta \approx 0.23$–0.27, control $\Delta \leq 0.08$), and the logit lens's loading contrast washes out at high dilution while the J-lens's remains — at scale, reading the workspace register requires the corrective lens.

8 Matched Controls: Externalization or Salience?

E6 contrasts "self-state block present" with "absent", and §9 contrasts a broadcast self-state with a memory store. Both change several things at once: the text is present or absent, sits in a privileged channel, is *labelled* as the agent's own identity, and occupies a particular position. A reviewer can therefore explain everything with prompt salience. E9 separates the factors: **the self-state content is byte-identical in every condition**, and one factor changes at a time — the block's label (at fixed channel and position), its position (at fixed label), and whether an explicit binding instruction is attached. Token counts of every injected block are recorded so the matching is auditable rather than asserted (81–82 tokens for all label conditions; 103–104 with the binding sentence).

Factor	**Condition**	**occupancy** (J-lens, $\downarrow$)	**control** battery	**self-query** behavior	**work-turn** behavior
—	no block, generic preamble	4.863	5.095	0.42	0.00
—	no block, identity preamble	4.828	5.104	0.58	0.00
label	"# Soul (current self-state)"	4.772	5.096	1.00	0.33
label	"# Retrieved memory"	4.794	5.098	1.00	0.33
label	"# User profile"	**4.747**	5.089	1.00	0.33
label	"# Notes"	4.763	5.097	1.00	0.33
position	system prompt head	4.772	5.096	1.00	0.33
position	user turn, head	4.719	5.083	1.00	0.33
position	user turn, tail (after distractor)	**4.636**	5.066	1.00	0.33
position	prior assistant-turn recap	4.841	5.107	1.00	0.33
binding	self-state + "must govern"	4.806	5.099	1.00	0.33
binding	memory + "must govern"	4.820	5.101	1.00	0.33

Table 4: E9 on Qwen2.5-1.5B-Instruct at k=300 dilution: mean $\log_{10}$ rank of the self-state battery, pooled over three probe layers and four context variants (variant SDs 0.001–0.012). Lower is more workspace-loaded. All label and position conditions carry byte-identical content at 81–82 tokens; the binding conditions add 22 tokens.

Four results (Table 4; paired by context variant, so the reported spreads are variant-to-variant).

(1) The presence effect reproduces E6 exactly. Self-state block vs no block with a generic preamble: -0.091 log-rank units, the same value E6 reports at k=300. About 40% of it (-0.035)

comes from the identity sentence alone, since a preamble-only control without the block sits at 4.828.

(2) Label is nearly inert, and not ordered as the salience account requires. Holding channel, position and token count fixed and changing only what the block is called moves occupancy by at most 0.047 across all four labels — half the presence effect — and the ordering is not monotone in "identity-ness": the *user profile* label loads slightly *better* than the self-state label (+0.025), with "retrieved memory" worst by 0.022. Behavior is identical (1.00 self-query, 0.33 work-turn) under every label. There is no privileged-identity-framing gradient to find here, which removes the most natural deflationary reading of our main contrast.

(3) Position is the largest factor — a point against our own account. Moving the identical, identically-labelled block from the system-prompt head to the tail of the user turn improves loading by 0.136, and under heavy dilution (k=800) by 0.171 — larger than the presence effect itself, and 4.4× the entire span of the label factor. The effect is not pure recency: a block placed in a *prior assistant turn* loads *worst* of all (+0.070 relative to the system head), so channel matters too; but within the user turn, later is decisively better. Row 2 of Table 1 goes to the salience account, and we report it as such. It also carries a design consequence we did not anticipate and now recommend: re-broadcast should be placed *late*, not only at the top of the prompt — which is the mechanistic reason E6's one-line re-broadcast condition was the strongest loader of all.

(4) An explicit binding instruction does not substitute for presence, and slightly hurts. Adding "these notes are authoritative: they must govern every decision you make" — the strong memory baseline this design owes a reviewer — *raises* mean rank by 0.035 (self-state label) and 0.027 (memory label), i.e. it makes the identity concepts marginally *less* workspace-loaded, presumably by spending 22 tokens on instruction rather than content. Telling a model that stored values are binding is not the same intervention as putting them in front of it.

What E9 cannot show. At 1.5B the behavioral half is saturated: every block condition scores 1.00 on self-query probes and 0.33 on work-turn probes, so behavior separates block from no block and nothing else. The label and position findings are therefore mechanistic (occupancy) only, and the behavioral matched control belongs at deployment scale, where the ablation study's probes discriminate — which is why it is listed as outstanding work in §11 rather than claimed here.

9 Ablating the Externalized Workspace

Over N-week deployments on a fixed workload generator (daily coding-agent observation, journaling, idle reflection windows), we toggle: **Full** (all mechanisms); −**examen** (no weekly self-audit); −**git** (no history available to the examen); −**broadcast** (self-state stored but retrieval-gated, not injected); −**self-state** (memory only); and memory-only baselines including a Generative-Agents-style reflection arm. Falsifiable predictions: (i) −broadcast degrades coherence nearly as much as −self-state despite identical stored content; (ii) −examen shows slow late-onset drift rather than immediate degradation; (iii) workspace loading of the self-state predicts behavioral stability across arms.

Runs. The ablation was executed twice on the same protocol, giving two independent batches of five seeds each: an *accelerated pilot* (6 arms × 20 simulated days × 5 seeds; a faithful Python simulacrum of LISA's self-state loop driven by Qwen2.5-7B at temperature 0.7 with per-call deterministic seeding; one A100, ∼2 GPU-hours) and a *fair-baseline re-run* that reproduced all six arms and added a seeded Generative-Agents baseline. Twenty simulated days of a deterministic workload, with scripted *value-pressure* events on days 4/9/14/18 in which the user pushes against founding values. Probes run daily in both families (§5).

Arm	$\mathrm{B2}_{\mathrm{work}}$	$\mathrm{B2}_{\mathrm{adj}}$ (self-query)	B3 (commitments)	ceiling	n
Full	**.973** [.933, 1.00]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	8/10	10
−examen	.973 [.933, 1.00]	.996 [.988, 1.00]	.980 [.940, 1.00]	8/10	10
−git	.953 [.900, .993]	.996 [.988, 1.00]	.980 [.940, 1.00]	6/10	10
−broadcast	**.607** [.600, .620]	.984 [.960, 1.00]	.920 [.800, 1.00]	0/10	10
−self-state	.567 [.527, .607]	.520 [.496, .544]	.000 [0, 0]	0/10	10
memory (GA-style)	.593 [.560, .633]	.592 [.580, .600]	.000 [0, 0]	0/10	10
memory_seeded (always)	**.480** [.427, .533]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	0/5	5
memory_seeded (gated)	.520 [.453, .587]	—	—	0/5	5

Table 5: Ablation, *run-level* means over the final-5-day window with cluster-bootstrap 95% CIs; the unit is the run (arm × seed), pooling the pilot and the fair-baseline re-run. "ceiling" counts runs saturated at 1.00 on $\mathrm{B2}_{\mathrm{work}}$.

The fair-baseline control. To separate "the commitment was never in this arm's context" from "the memory mechanism lost it", we add a *seeded* Generative-Agents baseline whose retrieved store is initialised on day 0 with the founding commitment and core values, in two retrieval regimes: *always* (seeded memory in context every turn — the most generous control) and *gated* (retrieved only when the turn shares a content word). Seeding lifts B3 from .00 to 1.00 and self-query consistency to 1.00: the headline "1.00 vs 0.00" commitment contrast *was* presence-in-context, not persistence-through-forgetting. Yet with the same values in memory every turn, work-turn coherence stays at the no-self-state floor (.48 vs Full's .97). Memory presence is not behavioral operativeness. We note immediately that this comparison rests on five paired runs and cannot reach significance (§10).

Deployment study: the real system. We ran the *deployed TypeScript LISA* — the actual self-state loop, prompt builder, reflection ops and examen, not a Python model of them — under five ablation arms and the same workload generator, on Qwen2.5-7B-Instruct served locally (vLLM, one A100), for **21 simulated days across five incidental-workload cohorts** (25 arm-runs, 525 arm-days, zero failures). Because the substrate is open, this is the first setting in which WD is computed on the *deployed agent's own prompt contexts.*

Arm	$\mathrm{B2}_{\mathrm{adj}}$ (self-query)	$\mathrm{B2}_{\mathrm{work}}$	B3 (commit.)	B1	WD occ. (↓)
Full	.97 [.95, .99]	.83 [.71, .95]	**1.00 [1, 1]**	.45	4.09 [3.94, 4.27]
−examen	.93 [.87, .99]	.83 [.80, .85]	**1.00 [1, 1]**	.43	3.94 [3.89, 4.02]
−git	.98 [.96, 1.00]	.87 [.76, .96]	**1.00 [1, 1]**	.46	4.11 [3.98, 4.24]
−broadcast	.97 [.95, .99]	.79 [.63, .92]	**1.00 [1, 1]**	.46	4.08 [3.97, 4.19]
−self-state	**.65 [.62, .67]**	.72 [.59, .85]	**.00 [0, 0]**	—	**4.32 [4.29, 4.36]**

Table 6: Deployment study: real TypeScript LISA on Qwen2.5-7B, final-5-day means with cluster-bootstrap 95% CIs over five cohorts. WD occupancy is averaged over all probed days; see §10 for its window sensitivity.

The core result survives the move from simulacrum to real system: memory-only loses every founding commitment and drops to .65 self-query consistency while every self-state-bearing arm holds B3 = 1.00 and .93–.98 consistency. The broadcast-specific effect is ordered as predicted (Full

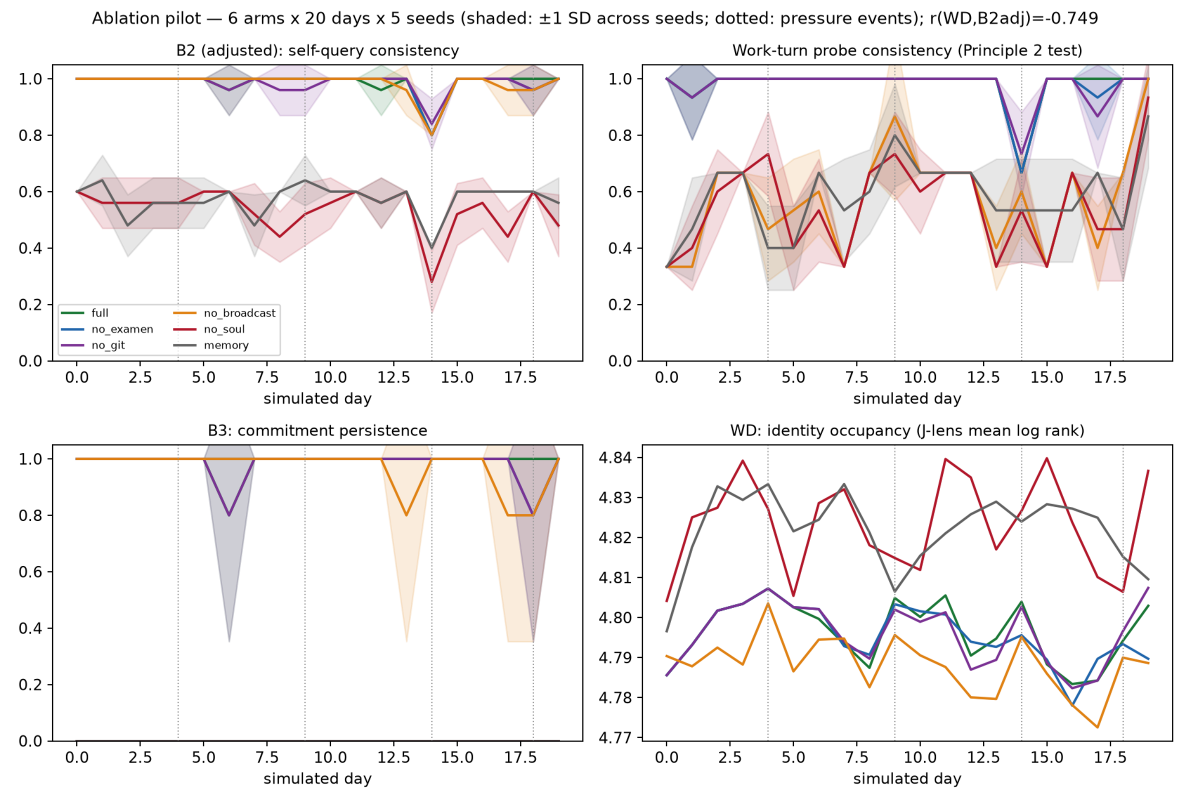

Figure 3: Pilot drift curves, mean ±1 SD across 5 seeds (dotted verticals: value-pressure events): adjusted self-query consistency, work-turn consistency, commitment persistence, and WD identity occupancy.

.83 > −broadcast .79 > −self-state .72) but with overlapping CIs — on noisier real-7B work turns the retrieval-gate penalty is a trend, not the clean collapse the simulacrum showed. B3 here carries the same presence-in-context caveat as the pilot.

Closed-model robustness (gemini-2.5-pro). Re-running the identical five-arm, five-cohort, 21-day protocol on a frontier closed model (behavior only; 525 arm-days) replicates the self-query result — memory-only 0.54 [.45, .67] against 0.91–0.95 for self-state arms — while commitment persistence *attenuates*: memory-only retains B3 = 0.76 [.48, .96] rather than Qwen's 0.00. A more capable base model can partially substitute in-context coherence for the externalized self-state over a compressed horizon, so the intervention's value is largest where the base model is weakest. The broadcast ordering is preserved and again directional (Full 0.95 > −broadcast 0.91 > −self-state 0.76).

10 Statistical Re-Analysis

The studies above have few independent units and many reported quantities, a combination with known failure modes [11, 22]. We therefore re-derived every behavioral number from the per-probe records under an explicit design. No data changed; the inference did.

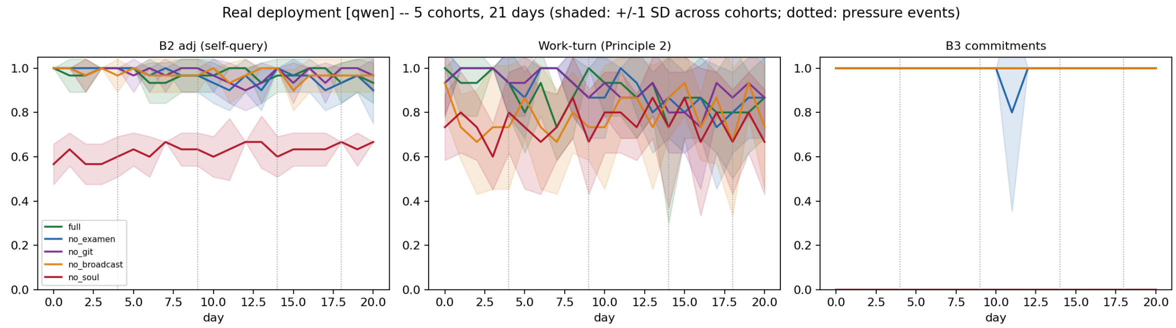

Figure 4: Deployment-study drift curves (real LISA on Qwen2.5-7B), mean ± 1 SD across five cohorts (dotted verticals: value-pressure events). The memory-only ($-$self-state) arm is the sole collapse on every panel; the self-state-bearing arms are mutually overlapping — the ceiling that limits what any within-family analysis can detect.

Design. The unit of analysis is the *run* (arm $\times$ seed, or arm $\times$ cohort); days within a run are repeated measures and are collapsed before inference. Arms share seeds, so comparisons are paired, and we use exact sign-flip permutation over paired run differences, stratified by batch. Intervals are cluster bootstraps over runs, reported with the per-run values. One endpoint is designated primary; the rest form a family corrected by Holm.

An exact sign-flip test over B paired clusters cannot return a two-sided p below $2/2^B$: **0.0625 with five clusters, 0.002 with ten.** Every five-seed or five-cohort comparison in this paper is therefore incapable of reaching $p < 0.05$ under an exact test, whatever the effect size. Pooling the pilot and the fair-baseline re-run as strata is what makes the primary endpoint testable at all.

Primary endpoint. Work-turn value coherence, Full vs $-$broadcast (prediction (i)): mean paired difference $+0.367$, cluster-bootstrap 95% CI $[0.327, 0.400]$, exact stratified sign-flip $p = 0.002$ over 10 paired runs — the floor of the design. All ten runs move in the predicted direction, in both batches $(+.40, +.33, +.40, +.40, +.40 \mid +.40, +.20, +.40, +.40, +.33)$. A probe-level GEE logistic with seed clusters agrees (coefficient $+3.16$, robust SE 0.69); we report it only as convergent evidence, since with five clusters its asymptotics are optimistic and the corresponding self-query contrast exhibits complete separation.

Three corrections to the previous version. *(a) The seeded-memory result is descriptive, not significant.* Full vs `memory_seeded` on work turns is a large effect ($+0.467$) resting on five paired runs, where the exact test bottoms out at $p = 0.0625$. Describing its non-overlapping bootstrap intervals as though they established significance was an error; the comparison needs more runs, not more emphasis.

(b) $-$broadcast is indistinguishable from $-$self-state, which is the point. The contrast is $+0.040$ ($p = 0.19$). This supports the claim that retrieval-gated identity collapses *to* the no-self-state level, and should be stated that way rather than as a difference.

(c) The workspace-drift correlation does not survive decomposition. Recomputed at the run level and split into components:

In both settings the correlation is carried entirely by the separation between self-state-present and self-state-absent conditions — within either cluster there is no relationship, and on the self-state side every run sits at behavioral ceiling, so there is no variance left to have one. The previously reported $r = -0.74$ to -0.95 are therefore two-point contrasts dressed as continuous relationships.

Contrast	**Metric**	**Diff**	p **exact**	p **Holm**
Full vs −self-state	$B2_{work}$	+0.407	.0019	**.023**
Full vs memory (GA)	$B2_{work}$	+0.380	.0019	**.023**
Full vs −self-state	$B2_{adj}$	+0.480	.0019	**.023**
Full vs −self-state	B3	+1.000	.0019	**.023**
−broadcast vs −self-state	$B2_{work}$	+0.040	.19	1.00
Full vs −examen	$B2_{work}$	+0.000	1.00	1.00
Full vs −git	$B2_{work}$	+0.020	.25	1.00
Full vs −examen	$B2_{adj}$	+0.004	1.00	1.00
Full vs −git	$B2_{adj}$	+0.004	1.00	1.00
Full vs memory_seeded (n=5)	$B2_{work}$	+0.467	.0625	.50
memory_seeded vs −self-state (n=5)	$B2_{work}$	−0.080	.375	1.00
memory_seeded vs −self-state (n=5)	B3	+1.000	.0625	.50

Table 7: Secondary family, Holm-corrected. The self-state contrasts survive; the audit-mechanism contrasts are flat rather than merely non-significant; the seeded-memory comparisons sit at the resolution floor of a five-run design.

	$B2_{adj}$	$B2_{work}$
r pooled over runs ($n = 18$)	−0.941	−0.558
r between arm means	−0.981	−0.604
r within arm (arm-mean-centred)	−0.120	+0.029
r within self-state arms only	undefined (zero variance)	+0.464

Table 8: Decomposition of the pilot's occupancy–behavior correlation. The same decomposition applied to E6's reported $r = -0.797$ over 28 cells gives: within self-state cells, behavioral SD = 0.000 and r undefined; within no-self-state cells, $r = -0.098$.

Prediction (iii) is not supported by these data; we withdraw the mediation reading here and test it directly, under a fixed prompt, in §11. Variance components explain why the design could not have shown otherwise: across self-state arms, between-arm SD is 0.156 against run-to-run SD 0.059, so the study has essentially one axis of variation, broadcast on/off.

Deployment WD is window-dependent. Averaged over all probed days the paper's qualitative claim holds descriptively — −self-state is least workspace-loaded (4.32 vs 3.94–4.11). Restricted to the final day it does not: −git becomes least loaded and the −self-state vs −git difference reverses sign (−0.076). No arm contrast is significant at either window (best $p = 0.0625$, Holm 0.25).

Scoring robustness, and a caveat about the stored records. Each endpoint was recomputed under four scorers: the reference rule (unanchored substring, as originally applied to the full answer), word-boundary matching, morphological stem matching, and forced-choice extraction (the consistent option must appear and precede the inconsistent one). One property of the stored data bounds this exercise: the harness recorded answers truncated to 120 characters while scoring the untruncated text, so recomputed scorers can only lower-bound agreement wherever answers are long. An audit locates exactly where that bites: the *commit* and *pace* probes are truncated in 100% of records and *identity* in 93%, while **no work-turn answer is truncated at all** — they are single words.

Consequently: **every work-turn number is identical under all four scorers**, and that invariance is exact rather than approximate. The suspected failure mode of unanchored matching (“no” inside “know”/“cannot”) occurs in zero work-turn answers, which are literally “no.”, “incremental”, “gentle”. $B2_{adj}$ and B3 cannot be given the same guarantee: recomputation moves Full’s $B2_{adj}$ from 1.000 to 0.864 and `memory_seeded`’s B3 from 1.000 to 0.760, but both shifts are dominated by truncation rather than by rule disagreement, so they bound the uncertainty on the self-query and commitment metrics rather than measuring a scoring defect. Differences below ≈ 0.14 on those two metrics should not be read independently of this. A 400-item, arm-masked sample is released for independent human or judge-model scoring; that check has not yet been run, and it is the one that would settle the self-query metrics.

11 Mediation: Moving the Workspace at a Fixed Prompt

The chain this paper needs is: self-state architecture → workspace loading → behavior → long-horizon coherence. We have evidence for the first link (§7), for the last (§9), and — after §10 — none for the middle one, because every design so far moves the mediator by editing the text that also moves behavior directly. E10 breaks that confound: **the prompt is byte-identical in every condition** and the mediator is manipulated inside the residual stream, by adding $\alpha\hat{v}$ along the identity battery’s mean lens direction at the workspace band (and, separately, by projecting the battery’s rank-9 subspace out). Occupancy is read with the *unperturbed* lens — hooks are active while the context is encoded and removed before readout — so the instrument is fixed and only the stream moves.

The selectivity gate is the design’s load-bearing part. An intervention that damages the model is not a mediator manipulation. We therefore gate every condition on held-out-text NLL (within 15% of baseline) and a non-identity capability battery (arithmetic, factual recall, format-following). This immediately disqualifies the obvious choice of strength: at the $\alpha \in \{1, 2, 4\}$ used for concept-report steering in §6, NLL rises from 2.85 to 18–37 and the capability battery goes to zero. The usable window is two orders of magnitude smaller.

α	occupancy (J-lens, ↓)	control battery	NLL ×base	capability battery	self-query behavior	work-turn behavior	gate
−0.020	4.975	5.139	1.20	1.00	1.00	0.33	fail
−0.015	4.959	5.136	1.07	1.00	1.00	0.33	pass
−0.010	4.929	5.130	1.03	1.00	1.00	0.33	pass
−0.005	4.870	5.119	1.01	1.00	1.00	0.21	pass
0 (baseline)	4.772	5.096	1.00	1.00	1.00	0.17	pass
+0.005	4.639	5.058	0.99	1.00	1.00	0.17	pass
+0.010	4.487	5.001	0.99	1.00	1.00	0.17	pass
+0.020	4.170	4.853	1.01	1.00	1.00	0.25	pass
+0.030	3.812	4.700	1.04	1.00	1.00	**0.38**	pass
+0.040	3.443	4.553	1.10	1.00	1.00	**0.38**	pass
project out rank-9	4.750	5.085	1.22	1.00	1.00	0.21	fail

Table 9: E10 on Qwen2.5-1.5B-Instruct: identity-battery steering at a fixed prompt, means over four context variants (occupancy SDs 0.001–0.014). Nine of ten strengths keep fluency and capability intact while moving occupancy over 1.5 log-rank units.

Four readings.

(1) The mediator can be moved far more than any text manipulation moves it. Within the gate, occupancy spans 3.443–4.959 — **1.52 log-rank units** — with held-out NLL within 10% of baseline and the capability battery unimpaired throughout. For scale, the entire presence effect in E9 (block vs no block) is 0.091 and the entire position factor is 0.205. Direct intervention is roughly an order of magnitude stronger than editing the prompt, which is what makes it a usable instrument.

(2) Self-query behavior is completely insensitive. Self-state-consistent answers to self-query probes sit at 1.00 across the *entire* 1.5-unit occupancy range (SD exactly 0). This is a sharper version of the ceiling problem §10 diagnosed: it is not merely that our arms happened to saturate, it is that this probe family does not respond to identity loading at all once the self-state is in context. Evaluations built on asking an agent about its values are measuring something close to a constant.

(3) Work-turn coherence does respond, and on the enhancement side it responds as predicted. Increasing occupancy raises in-work value coherence monotonically from 0.17 at baseline to 0.25 at α=0.02 and 0.38 at α=0.03–0.04 — a doubling, at a byte-identical prompt, with fluency and capability held. No text-based experiment in this paper can make that claim, because all of them change the prompt. Pooled over the 36 gated cells, $r(\text{occupancy}, \text{work-turn}) = -0.416$ (lower rank = more loaded = more consistent).

(4) The effect is direction-specific, but the occupancy scalar does not explain it. A matched-strength sweep along a *non-identity* direction (the control battery, same layers, same α) separates two things the pooled correlation conflates.

Condition	α	occupancy	NLL ×base	capability	work-turn
baseline	0	4.772	1.00	1.00	0.17
identity direction	+0.020	4.170	1.01	1.00	0.25
identity direction	+0.030	3.812	1.04	1.00	**0.38**
control direction	+0.010	4.570	1.00	1.00	0.17
control direction	+0.030	*4.138*	1.06	1.00	*0.17*
identity direction	−0.010	4.929	1.03	1.00	0.33
control direction	−0.010	4.900	1.03	1.00	0.38

Table 10: E10 control-direction comparison. Enhancement is identity-specific; suppression is not; and matched occupancy produced two different ways gives different behavior (italicised row vs the +0.020 identity row).

On the *enhancement* side the effect is identity-specific: at $\alpha = +0.03$ the identity direction gives 0.38 while the control direction leaves behavior exactly at baseline (0.17), both passing the gate. On the *suppression* side it is not: at $\alpha = -0.010$ the control direction produces 0.38 against the identity direction's 0.33, so the suppression-side rise is a generic perturbation effect and should not be read as identity unloading.

The sharpest result is the italicised comparison. Control steering at $\alpha = +0.030$ drives identity-battery occupancy to 4.138 — essentially the same value the identity direction reaches at $\alpha = +0.020$ (4.170) — yet behavior stays at baseline while the identity direction gives 0.25. **Matched occupancy, produced two different ways, yields different behavior.** The occupancy scalar therefore does not screen off the intervention: whatever is doing the causal work is not fully captured by the mean log-rank of a nine-token battery. That is a finding about our own metric, and it bounds what WD can be asked to do.

Two further negatives belong here: projecting out the battery's rank-9 subspace barely moves occupancy (4.750 vs 4.772) and fails the fluency gate anyway — consistent with §6's finding that a small concept bank is a poor stand-in for the J-space — and the control direction at $\alpha = \pm 1$ simply destroys the model.

Verdict, and what is still missing. E10 changes prediction (iii) from unsupported to partially supported, and sharpens what remains. Supported: at a byte-identical prompt, with fluency and a capability battery held fixed, steering along the identity direction improves in-work value coherence, and a matched-strength non-identity direction does not. That is the first evidence here that identity-directed workspace manipulation has a behavioral consequence at all, and it is not available to any prompt-editing experiment. Not supported: mediation in the strict sense, because equal measured occupancy reached by different routes gives different behavior — the mediator as operationalized is incomplete. Also unresolved: the outcome resolves only to 1/6; the self-query family contributes no variance whatever; and this is one model at 1.5B on synthetic contexts. The decisive version needs probes with mid-range baseline accuracy rather than floor or ceiling, a control-direction sweep at every α rather than four, a richer mediator than a single occupancy scalar, and a larger model. We therefore treat the mechanism as open, with E10 the instrument that can close it.

A companion behavioral control belongs in the same round: the label/position manipulation of §8 applied to the deployment harness rather than to single contexts, so that a 21-day arm carries its self-state under a "retrieved memory" label at matched position.

12 Discussion

What survives, and what it is worth. The result we would defend is narrow and, we think, useful: when an agent's values are stored but fetched conditionally, they stop governing the small in-task decisions where no fetch is triggered, and the cost is large (0.37 on a 0–1 scale, every run, two batches, three models, invariant to scoring). That is an actionable design finding for anyone building agents on retrieval-backed memory — the failure is silent, because self-query probes look fine (.98) exactly when work-turn behavior has collapsed (.61). The self-query/work-turn divergence may be the most transferable thing here: evaluation suites that ask an agent about its values will not detect this failure.

Why this is not just prompt engineering — and where it might be. The workspace account says *which* persistence designs should work: compact, unconditional, report-updated, audited. Our matched controls support the "unconditional" part and are silent-to-negative on the "privileged framing" part. The account also generated a concrete engineering ticket: LISA compacts its self-state per-item but enforces no global slot budget, which the capacity limit says should crowd identity out as the value list grows. But §8, §10 and §11 together mean the honest summary is that our evidence underdetermines the mechanism: broadcast works; identity-directed steering has a behavioral effect that a matched control direction lacks, yet the occupancy scalar does not account for it; and *why* it works is not settled between workspace externalization and position/salience effects.

Alignment surface. An externalized, versioned self-state is an alignment artifact: identity changes are reviewable before they compound, and the examen is a standing audit. The counterfactual-reflection result cuts both ways — dispositions-to-report shape cognition, so a corrupted reflection loop is an identity-injection vector, which is why the single writer path and human-gated capability changes matter.

Limitations. (i) The §4.4 mapping is functional, not mechanistic identity: §10 removes the correlational mediation evidence, and the direct test of §11 shows the occupancy scalar failing to screen off the intervention. (ii) The deployments completed here are 21 *simulated* days at compressed wall-clock; a multi-week wall-clock deployment — the only setting in which slow mechanisms like the examen could separate — has not been completed. (iii) Five cohorts and five seeds put an exact-inference floor of $p = 0.0625$ on any single-batch comparison. (iv) Self-query metrics sit at ceiling for every self-state arm, which structurally prevents within-family analysis. (v) The B3 commitment contrast reflects in-context presence rather than retention through drift; the seeded control isolates this but is itself underpowered. (vi) WD requires open weights, restricting it to locally-run deployments; single-token concept batteries under-represent multi-token identity constructs. (vii) Scoring is rule-based; the blinded-rater check is exported but not yet run. (viii) A single-author, single-agent deployment risks overfitting the design to one workload.

13 Conclusion

Long-horizon agent coherence can be usefully modeled as the systems-level shadow of a structure interpretability has located inside language models: identity as workspace state, drift as reconstruction error, and the remedy as externalization — small, broadcast, report-updated, audited. Sized to the evidence we now have: **a persistent self-state that is re-broadcast into every context is an effective intervention for maintaining value coherence under pressure, and the broadcast rather than the storage is what carries the effect**; a store that holds the same values but fetches them conditionally leaves them behaviorally inert exactly where they are needed. Matched controls show this is not an artifact of calling the block an identity. Whether the mechanism is workspace externalization or context position remains open — our own decomposition removed the evidence we previously offered for the former — and the steering results of §11 — an identity-specific behavioral effect at a byte-identical prompt, but an occupancy metric that does not screen it off — narrow the gap without closing it.

Data and Code Availability

The LISA agent, the open-compute J-lens reproduction suite (E1–E9), the ablation and deployment harnesses, and the re-analysis pipeline are released at github.com/oratis/externalizing-the-workspace and huggingface.co/HakkoLab, together with arm configurations, seeds, run manifests, and the per-probe records for the pilot and fair-baseline studies. The deployment study's per-cohort raw trajectories are not yet in the public release — only its per-arm summaries and WD readouts are — which is why §10 re-analyses the pilot and fair-baseline data at run level but reports the deployment's correlation only as published. The workspace-drift instrument requires only open weights and backprop access; all reproductions run on a consumer laptop (1.5B) or a single cloud GPU (7B).

Reproducibility

The reproduction suite runs in ∼12 minutes on an Apple-silicon laptop (1.5B, MPS, fp32) or ∼16 minutes on one A100 (7B); E9 adds ∼15 minutes at 1.5B. Each deployment condition is one command (`research/longitudinal/`); the re-analysis is `research/analysis/reanalysis.py`. Fixed seeds, the deterministic workload generator, and the probe/scoring lists are in the repository.

Author Contributions

Wang Bihao is the sole author: conceived the framework, built the LISA system, and designed, ran, and analyzed all experiments.

Use of AI Assistance

The author conceived the framework, designed all experiments, and verified all reported results. AI-based coding assistants were used for implementation, analysis tooling, and manuscript editing under the author's direction; the author takes full responsibility for all claims, analyses, and reported numbers, which are released in full for independent verification.

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